

to fully plumb. So why couldn't it solve some high school-level equations now and then? More pressingly, the inner physicist theory is difficult to refute because it's hard to think of alternatives to it: if dogs can't solve partial differential equations to catch Frisbees, then how do they catch them anyway? They take great leaps into the air and catch erratically moving Frisbees in their jaws like it's nothing. If they aren't solving some physics problem in their brains, then how else could they (and we) possibly know how to precisely intercept a ball?

As recently as 1967, no one had a good answer. That year, the engineer Vannevar Bush wrote a book in which he described the scientific features of baseball as he understood them, and he was unable to provide any explanation for how outfielders know where to run to catch fly balls. Luckily for us, the physicist Seville Chapman read Bush's book and was inspired to propose a theory of his own the very next year.

The Algorithmic Approach

Chapman, true scientist that he was, was not satisfied with a mystical and unverified trust in the human subconscious, and he wanted a more concrete explanation for outfielders' powers. This is what he discovered.

Thinking with Your Neck

Chapman began to tackle the outfielder problem by noting the information available to someone catching a ball. Though it's difficult for humans to estimate an exact velocity or the trajectory of a parabolic arc, he thought we would have an easier time observing angles. If someone throws or hits a ball from the ground and the ground is flat and even, then the outfielder will see the ball start at close to eye level. Imagine an angle formed by two lines: the ground, and the line between the outfielder's eyes and the ball. The moment the ball is hit by the batter, this angle will be (roughly) 0 degrees. After the ball has been in flight for a brief moment, it will be higher than the ground, so the angle between the ground and the outfielder's line of sight with the ball will have increased. Even if the outfielder has not studied geometry, they will have a "feel" for this angle—for example, by feeling how far back they have to tilt their neck to see the ball.